Square Root of Strongly Convex and Strictly Positive need not be Convex

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At some stage, I had the idea that 'typically', the square root of a strongly convex and strictly positive function should be more-or-less convex. Whether or not this is true in some typical sense, it is certainly not literally true, as the following example demonstrates.

Let $\theta \in (0,1)$, and define $F_\theta(x) = \frac{1}{2} \cdot x^2 + \theta \cdot \cos x$. It is straightforward to check that for all $x \in \mathbf{R}$, $F_\theta(x) \geqslant F_\theta(0) = \theta > 0$ (i.e. the origin is a global minimiser of F_θ), and $\mathrm{D}^2 F_\theta \geqslant 1 - \theta > 0$, so F_θ is strongly convex and strictly positive.

Now, compute that

$$\begin{aligned} \mathrm{D}\left(F_\theta\right)^{1/2} &= \frac{1}{2} \cdot \left(F_\theta\right)^{-1/2} \cdot \mathrm{D}F_\theta \\ &= \frac{1}{2} \cdot \left(F_\theta\right)^{-1/2} \cdot \left(x - \theta \cdot \sin x\right), \end{aligned}$$

and

$$\begin{aligned} \mathrm{D}^2\left(F_\theta\right)^{1/2} &= \mathrm{D}\left\{\frac{1}{2} \cdot \left(F_\theta\right)^{-1/2} \cdot \left(x - \theta \cdot \sin x\right)\right\} \\ &= -\frac{1}{4} \cdot \left(F_\theta\right)^{-3/2} \cdot \left(x - \theta \cdot \sin x\right)^2 + \frac{1}{2} \cdot \left(F_\theta\right)^{-1/2} \cdot \left(1 - \theta \cdot \cos x\right) \\ &= -\frac{1}{4} \cdot \left(F_\theta\right)^{-3/2} \cdot \left\{\left(x - \theta \cdot \sin x\right)^2 - 2 \cdot F_\theta \cdot \left(1 - \theta \cdot \cos x\right)\right\}. \end{aligned}$$

This will be negative precisely when

$$\begin{aligned} \left(x - \theta \cdot \sin x\right)^2 &> 2 \cdot F_\theta \cdot \left(1 - \theta \cdot \cos x\right) \\ &= \left(x^2 + 2 \cdot \theta \cdot \cos x\right) \cdot \left(1 - \theta \cdot \cos x\right) \\ \iff \quad x^2 - 2 \cdot \theta \cdot x \cdot \sin x + \theta^2 \cdot \sin^2 x &> x^2 + \left(2 - x^2\right) \cdot \theta \cdot \cos x - 2 \cdot \theta^2 \cdot \cos^2 x \\ \iff \quad \theta \cdot \left(\sin^2 x + 2 \cdot \cos^2 x\right) &> 2 \cdot x \cdot \sin x + \left(2 - x^2\right) \cdot \cos x. \end{aligned}$$

Consider now values of x such that $x > 0$, $x^2 > 2$ and $\sin x < 0 < \cos x$, which occurs 'about a quarter of the time' for large x. For such values of x, it holds that

$$\theta \cdot \left(\sin^2 x + 2 \cdot \cos^2 x\right) = \theta \cdot \left(1 + \cos^2 x\right) \geqslant \theta > 0$$

and that

$$\begin{aligned} 2 \cdot x \cdot \sin x + \left(2 - x^2\right) \cdot \cos x &< 2 \cdot 0 + 0 \cdot 0 \\ &< 0 \end{aligned}$$

from which one deduces that $\mathrm{D}^2\left(F_\theta\right)^{1/2} < 0$ infinitely often. As such, the claim fails in a rather strong sense for this function.

Some visual checks can help with the intuitive picture here:
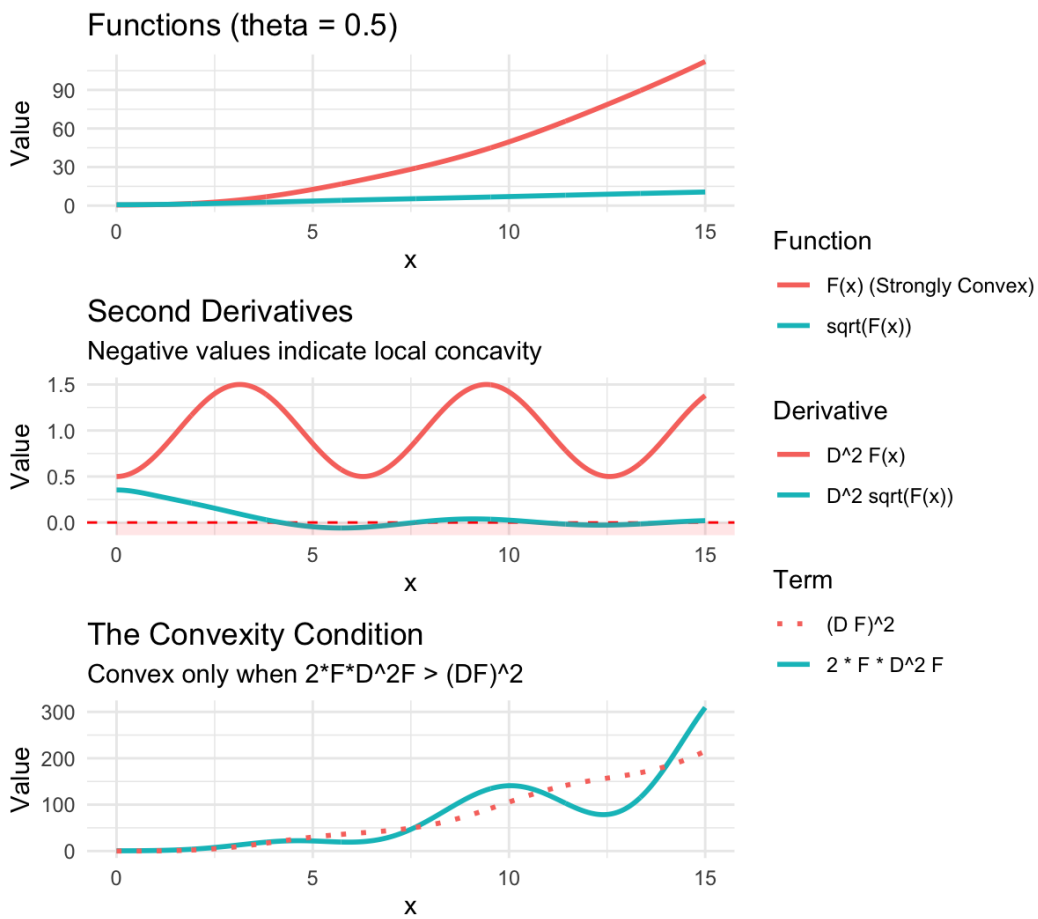
Figure 1: F_θ, $\mathrm{D}^2 F_\theta$, Convexity Condition